How Spark is Enabling the New Wave of Converged Applications

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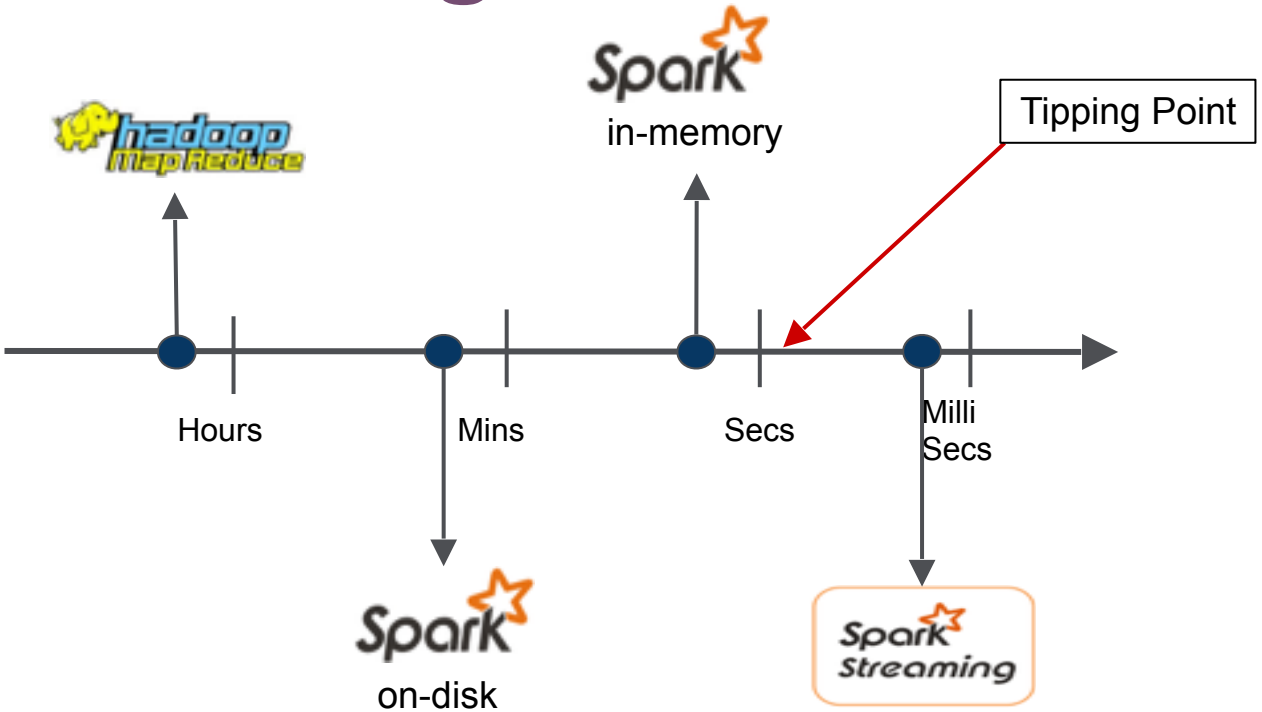
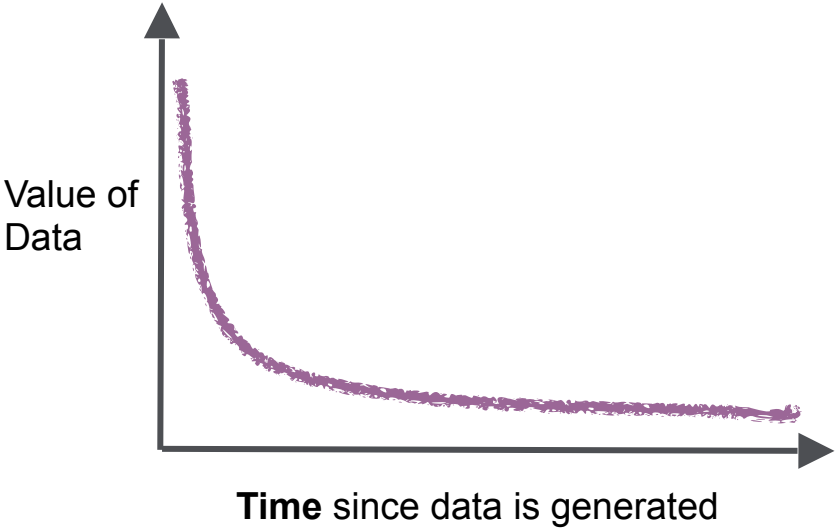
MapR Technologies

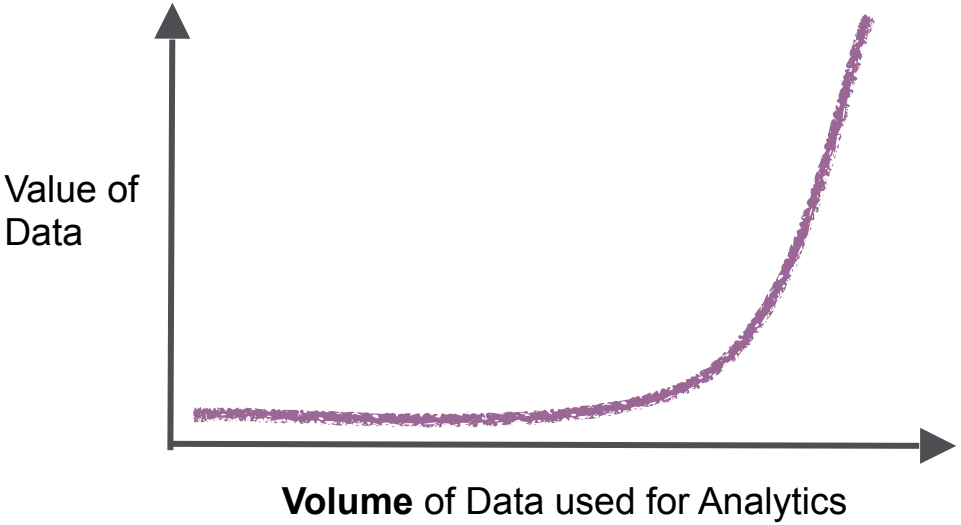
SPARK SUMMIT EUROPE 2016

Decreasing Job Latencies



Analytics & ETL: Batch or Continuous ?

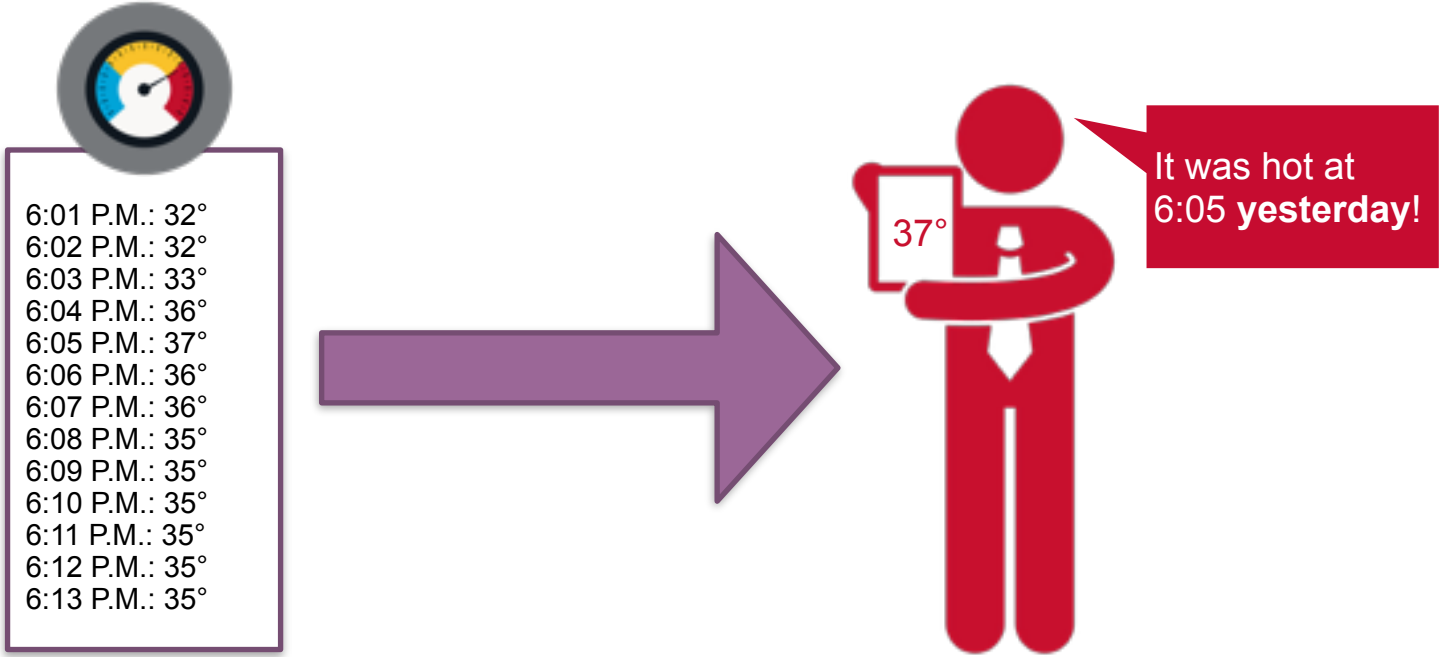




It's not an either or, **you have to do both**

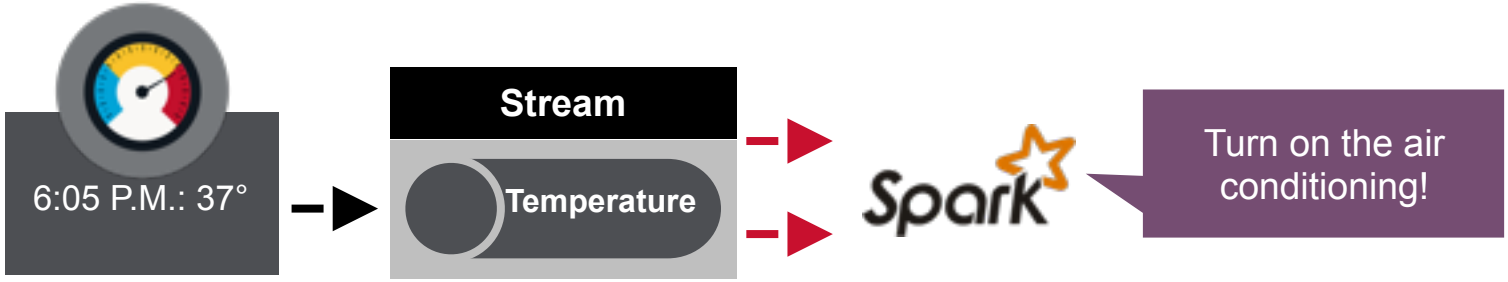
Why Stream Processing?

Batch processing may **be too late** for some events



Why Stream Processing?

It's becoming important to **process events as they arrive**



Advanced Analytics

Descriptive

- What Happened
- Why did it happen
- Discovery in nature
- **Batch** Analytics

Predictive

- What will happen
- Combines historical data with rules and algorithms
- **ML** (Batch + Real Time)

Streaming

- Analyse data as it happens
- Triggers and Alarms.
- Anomaly detection
- **Continuous** ETL and Analytics

Prescriptive

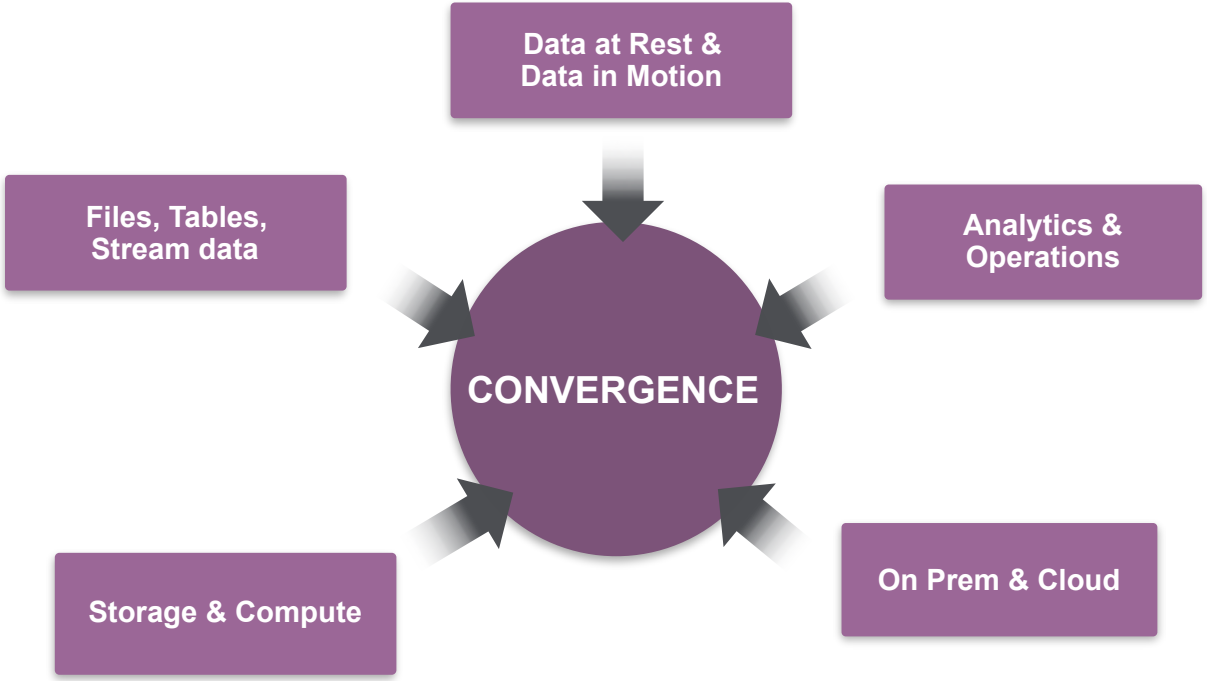
- What + When + Why
- Suggestions to take advantage of future opportunity or mitigate risks
- **Agility** is key to success.

There is a need to **converged** these Analytics

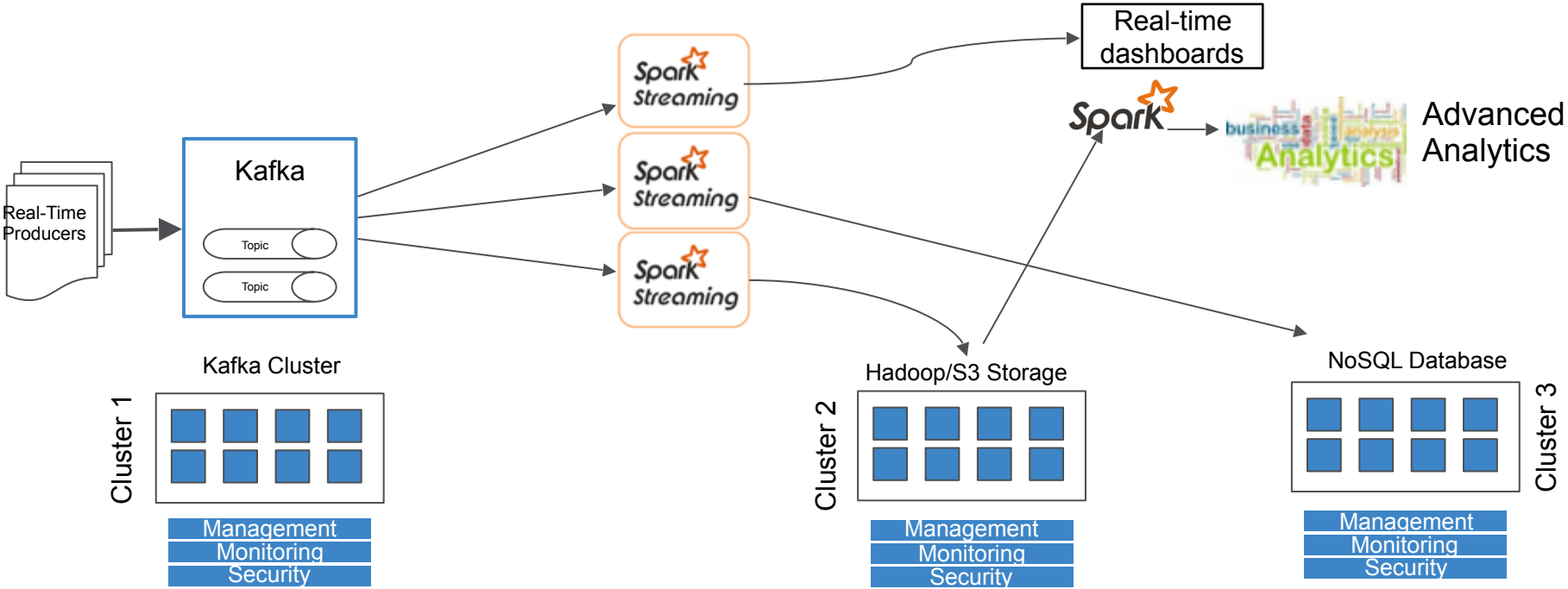
Converged Computing

	Offline	Real Time
Programmatic	Spark & ML	Spark Streaming
SQL	Spark SQL	Spark Structured Streaming

The Many “Convergences” In Progress

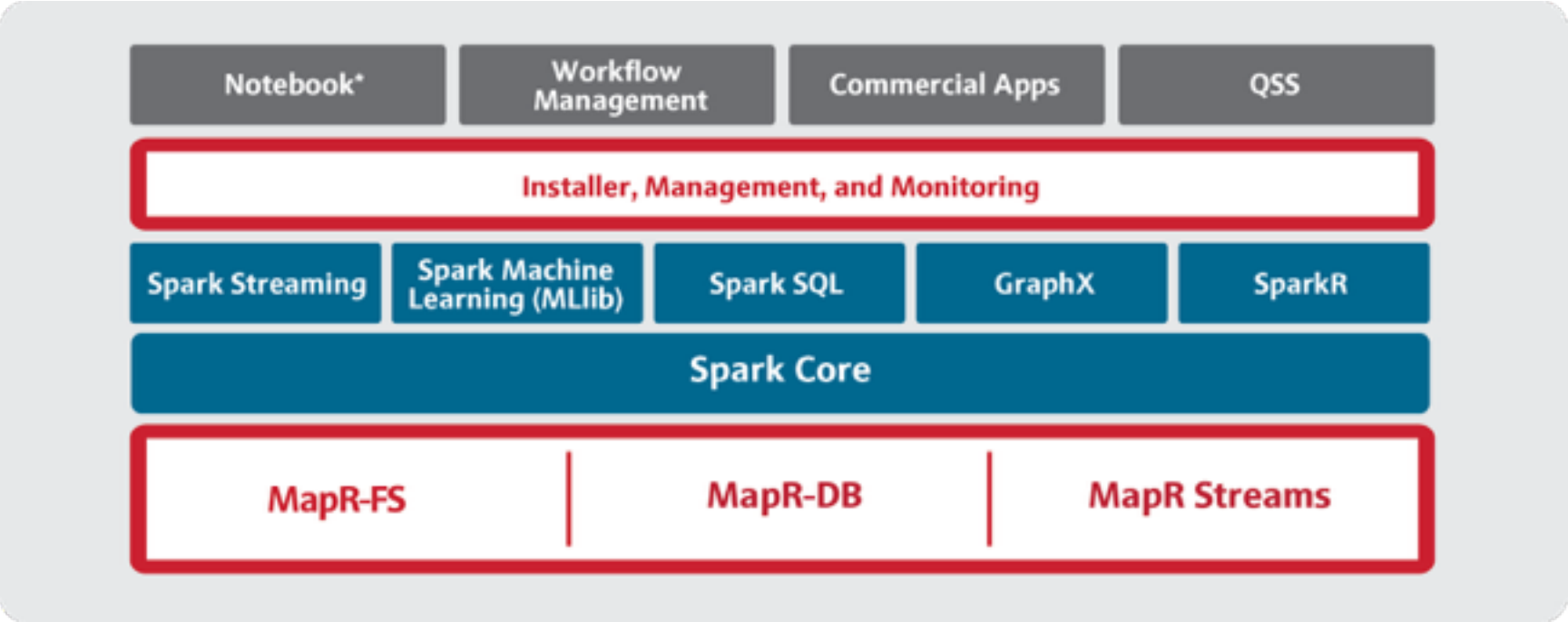


Spark on Non-Converged Platform



- Redundant 3x **Management, Monitoring and Security**
- Redundant 3x **Data Storage**

Converged Computing & Converged Data Management



MapR Converged Data Platform

Processing

Open Source Engines & Tools

- Hadoop
- Spark
- Apache Drill
- Search and Others

Commercial Engines & Applications

- Cloud and Managed Services
- Vertica, SAP, MySQL
- Custom Apps

Unified Management and Monitoring

Data

HDFS API | POSIX, NFS | HBase API | JSON API | Kafka API

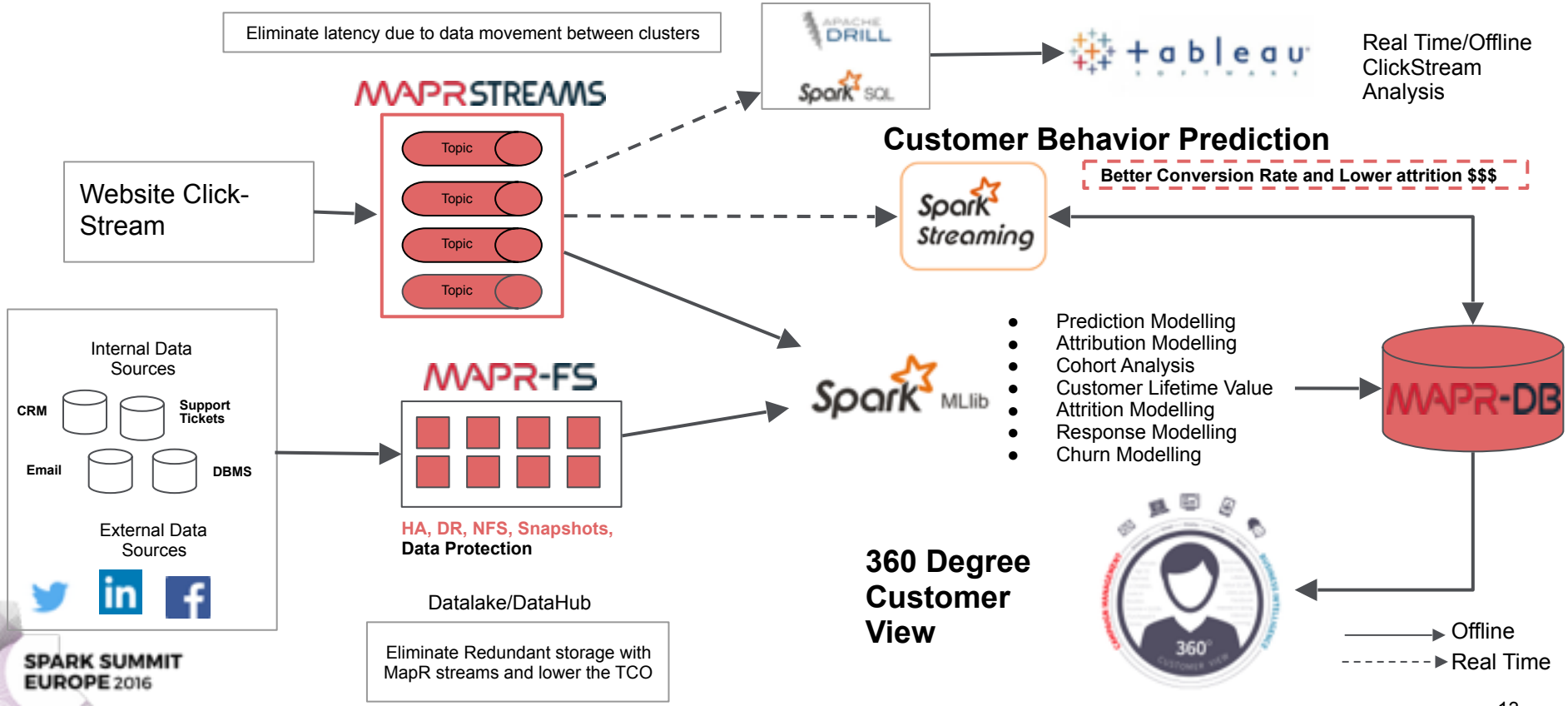
Web-Scale Storage	Database	Event Streaming
MapR-FS	MapR-DB	MapR Streams

High Availability | Real Time | Unified Security | Multi-tenancy | Disaster Recovery | Global Namespace

Enterprise-Grade Platform Services

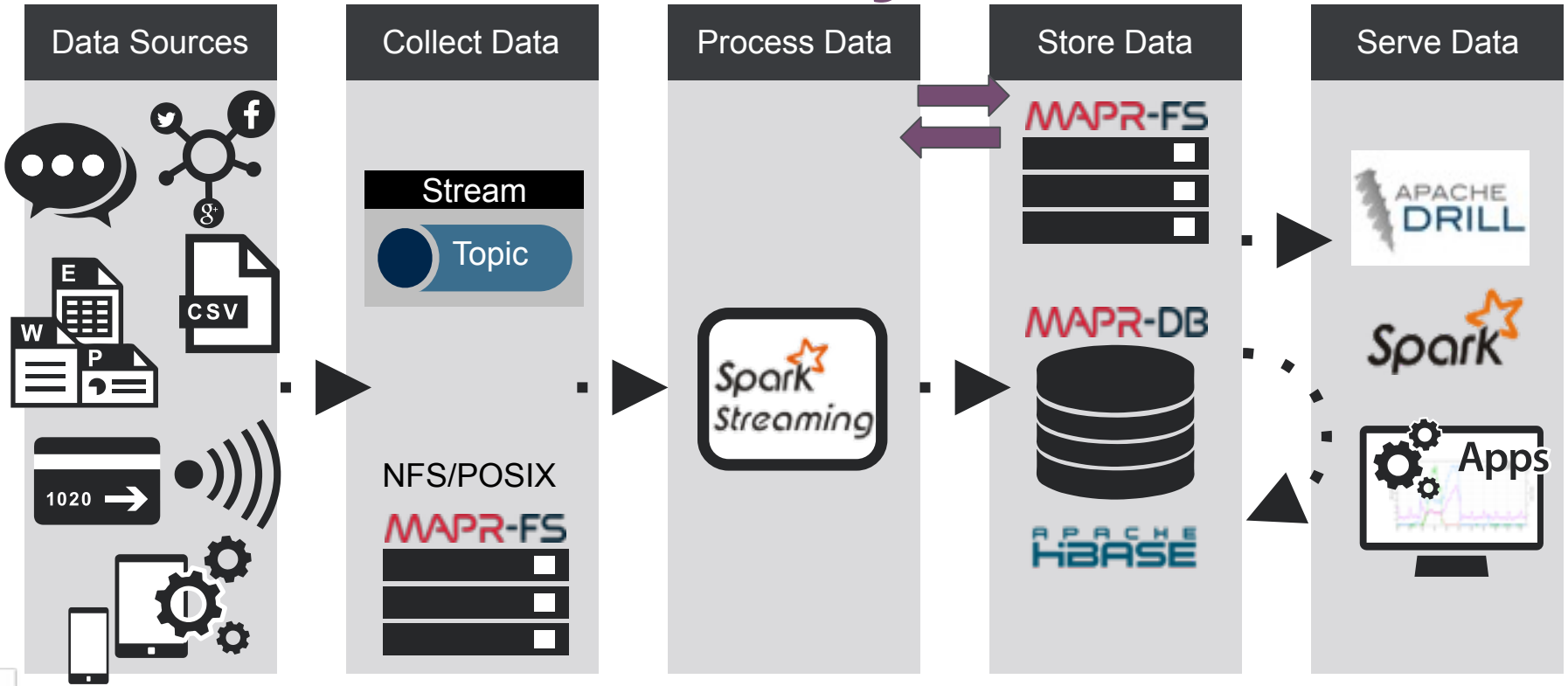
SAMPLE CUSTOMER USE CASE

Customer 360 & Behavior prediction

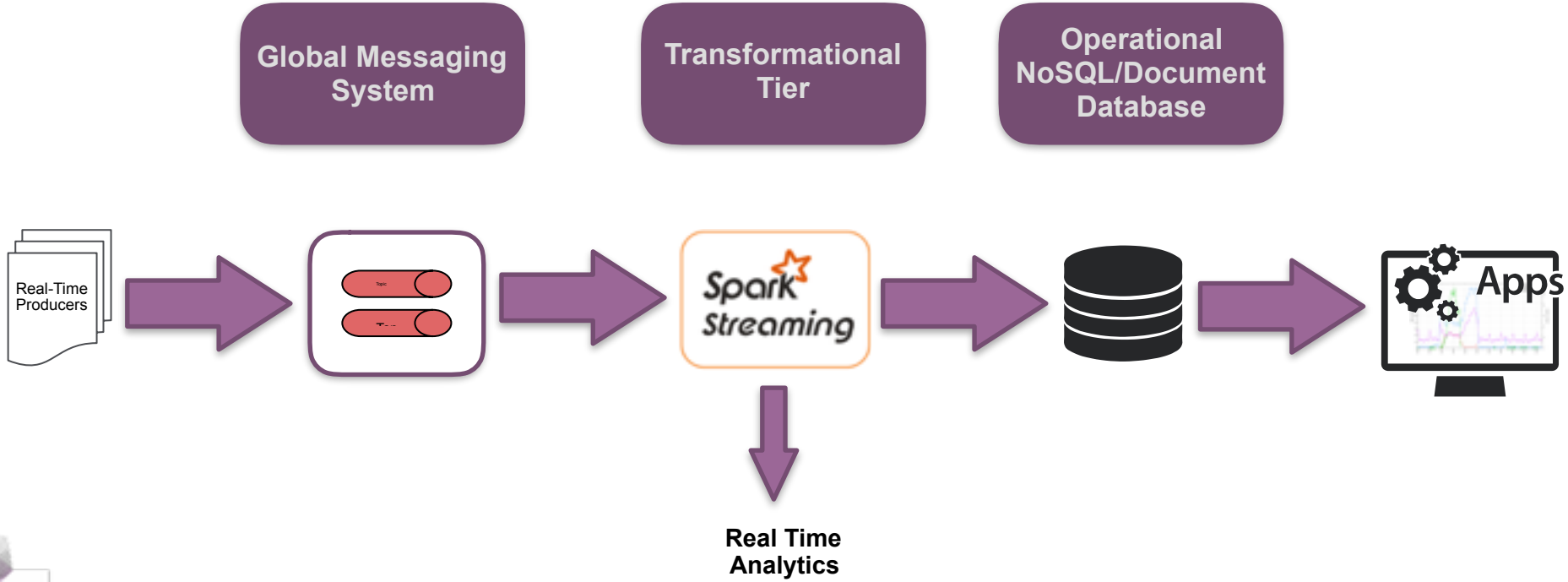


STREAMING FIRST ARCHITECTURE

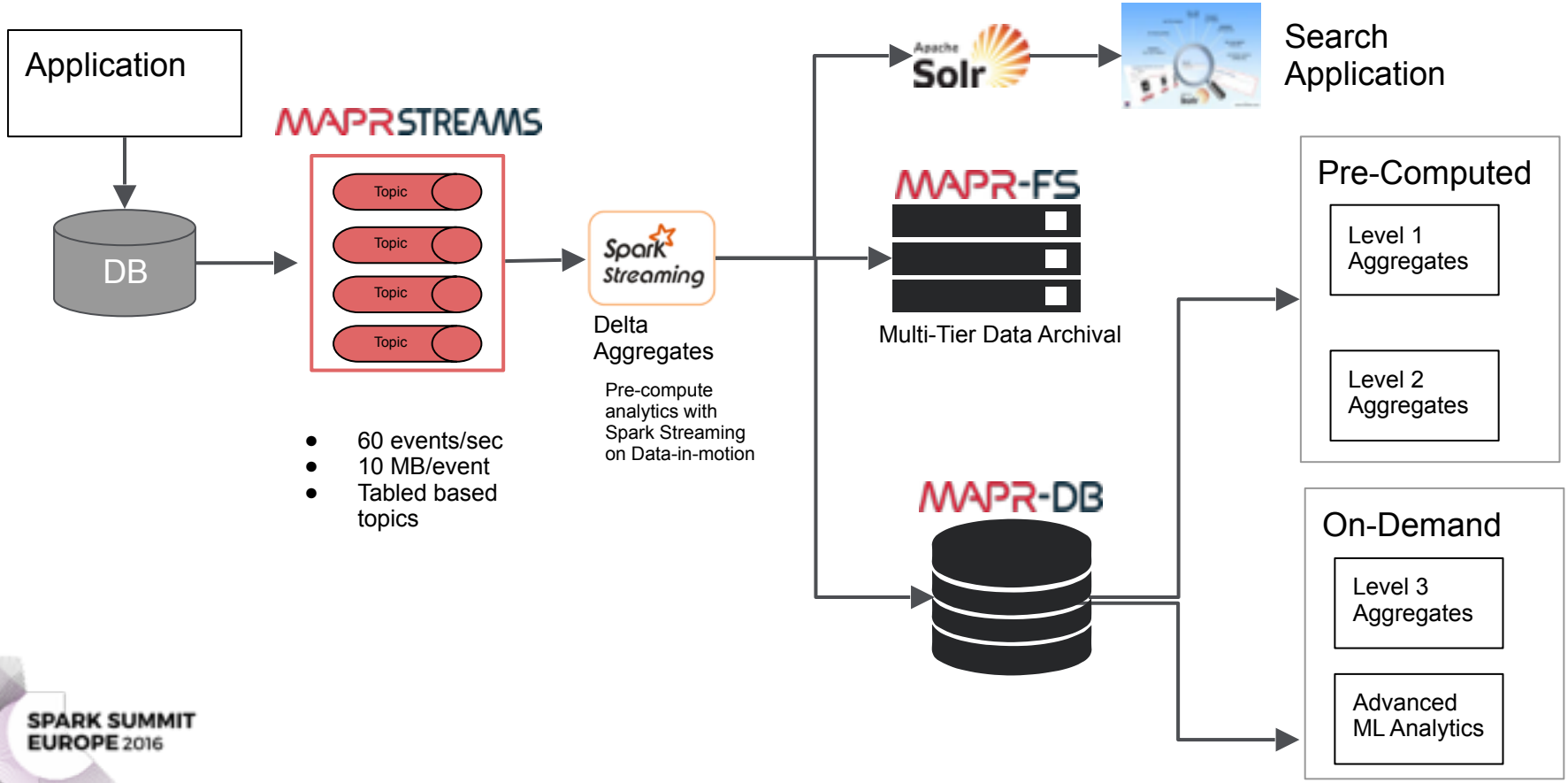
What Do We Exactly Need to Do ?



Trinity of Real Time



Continuous Streaming ETL & Computed Analytics



Q&A

Engage with us!

1. Read explanation of and Download code
   - https://www.mapr.com/blog/fast-scalable-streaming-applications-mapr-streams-spark-streaming-and-mapr-db
   - https://www.mapr.com/blog/spark-streaming-hbase

2. **Get Started:** MapR Converged Data Platform
   https://www.mapr.com/get-started-with-mapr

3. **Get Answers:** MapR Converge Community
   https://community.mapr.com/community/answers

4. **Get Trained**: MapR On-Demand Training
   https://learn.mapr.com

DEV 360 - Apache Spark Essentials

FREE

DEV 361 - Build and Monitor Apache Spark Applications

FREE

DEV 362 - Create Data Pipelines Using Apache Spark

FREE

THANK YOU.

Contact information or call to action goes here.

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